# Day 2: Java Programming Fundamentals

Study Notes and Practice Programs

## 1. Introduction to Java: JDK, JRE and JVM

Java is a general-purpose, object-oriented language. You write code once, compile it to bytecode, and run it on any machine that has a JVM ("write once, run anywhere").

| Component | What it is | Used for |
|---|---|---|
| **JVM** (Java Virtual Machine) | Runs bytecode (.class files) on your OS | Execution |
| **JRE** (Java Runtime Environment) | JVM + core libraries | Running Java programs |
| **JDK** (Java Development Kit) | JRE + compiler (javac) + tools | Writing and compiling programs |

**Flow:** Hello.java → javac → Hello.class (bytecode) → JVM → output.

## 2. Environment Setup

1. Download and install a JDK (version 17 or 21 LTS) from Adoptium (Temurin) or Oracle.
2. Set `JAVA_HOME` to the JDK folder and add its `bin` folder to `PATH` (the Windows installer often does this).
3. Verify in a terminal: `java -version` and `javac -version`.
4. Install an IDE: IntelliJ IDEA Community, Eclipse, or VS Code with the Java Extension Pack.
5. Test with Hello.java, shown below. Run it with `javac Hello.java` then `java Hello`.

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello, Java!");
    }
}
```

## 3. Core Concepts

- **Variables and data types:** primitives are byte, short, int, long, float, double, char and boolean. String is a class, not a primitive.
- **Operators:** arithmetic (+ - * / %), relational (== != < > <= >=), logical (&& || !), assignment (= += -=), increment/decrement (++ --).
- **Conditionals:** if, if-else, else if, switch.
- **Loops:** for, while, do-while, plus break and continue.
- **Input/output:** System.out.println() for output and Scanner for input.

## 4. Practice Programs

Each program needs `import java.util.Scanner;` at the top of the file.

### 4.1 Even or Odd

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        System.out.println(n % 2 == 0 ? n + " is even" : n + " is odd");
    }
}
```

### 4.2 Largest of Three Numbers

```
import java.util.Scanner;

public class Largest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
        int max = a;
        if (b > max) max = b;
        if (c > max) max = c;
        System.out.println("Largest: " + max);
    }
}
```

### 4.3 Factorial

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        long fact = 1;
        for (int i = 2; i <= n; i++) {
            fact *= i;
        }
        System.out.println(n + "! = " + fact);
    }
}
```

**Note:** long holds results up to 20!. Beyond that it overflows (use BigInteger).

### 4.4 Prime Check

```
import java.util.Scanner;

public class Prime {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        boolean isPrime = n > 1;
        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0) {
                isPrime = false;
                break;
            }
        }
        System.out.println(n + (isPrime ? " is prime" : " is not prime"));
    }
}
```

**Note:** checking only up to the square root of n is enough, because any larger factor pairs with a smaller one.

### 4.5 Reverse a Number

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int rev = 0;
        while (n != 0) {
            rev = rev * 10 + n % 10;  // take last digit, append it
            n /= 10;                  // drop last digit
        }
        System.out.println("Reversed: " + rev);
    }
}
```

### 4.6 Multiplication Table

```
import java.util.Scanner;

public class Table {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }
    }
}
```

## 5. Day 2 Checklist

[ ] JDK installed; java -version and javac -version work

[ ] IDE set up and Hello World runs

[ ] All six programs written, run, and tested with edge cases (0, 1, negatives, large values)

[ ] Can explain JDK vs JRE vs JVM in own words

## Stretch Exercises

- Prime check for every number in a range
- Palindrome number check (reuse the reverse logic)
- Fibonacci series
- Sum of digits
- Simple calculator using switch